

Lewis T. Kunik

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Education

- Ph.D. Atmospheric Sciences** 2026
University of Utah
Dissertation title: Monitoring indicators of photosynthesis in evergreen conifer forests with satellite-based optical remote sensing
- B.S. Computer Engineering & Computer Science** 2016
University of Wisconsin–Madison

Employment

- Postdoctoral Researcher**, University of Utah, Atmospheric Sciences 2026–present
- Graduate Research Assistant**, University of Utah, Atmospheric Sciences 2021–2026
- Air Quality Consultant**, Ramboll U.S. Corporation 2019–2021

Research and Professional Interests

- Applications of terrestrial remote sensing for mapping vegetation dynamics
- Large-dataset analytics for monitoring continental-scale land surface processes
- Climate change and human impacts on terrestrial ecosystems and biogeochemical cycles
- Understanding impacts of wildfire, drought, pest/pathogen using geostatistics and remote sensing
- Air quality monitoring in urban & natural environments using advanced computing techniques

Publications

Kunik, L., Bowling, D.R., Raczka, B., Hicke, J.A., Frankenberg, C., Cheng, R., Slaton, M.R., and Lin, J.C., 2026. Characterizing effects of tree mortality from wildfire and bark beetles using satellite observations of solar-induced chlorophyll fluorescence. *Remote Sens. Environ.*, 344, 115550. DOI: 10.1016/j.rse.2026.115550
Media coverage: University of Utah @theU Research Spotlight

Roten, D.R., Lin, J.C., **Kunik, L.**, Mallia, D.V., Wu, D., Oda, T., and Kort, E., 2024. The information content of dense carbon dioxide measurements from space: an urban-focused inversion approach with synthetic data from the OCO-3 instrument. *ESS Open Archive*. DOI: 10.22541/essoar.172485642.21372186/v1

Yang, J.C., Bowling, D.R., Smith, K.R., **Kunik, L.**, Raczka, B., Anderegg, W.R.L., Bahn, M., Blanken, P., Richardson, A.D., Burns, S.P., Bohrer, G., Desai, A.R., Altaf Arain, M., Staebler, R.M., Ouimette, P.A., Munger, J.W. and Litvak, M.E., 2024. Forest carbon uptake as influenced by snowpack and length of photosynthesis season in seasonally snow-covered forests of North America. *Agric. For. Meteorol.*, 353, 110054. DOI: 10.1016/j.agrformet.2024.110054

Kunik, L., Bowling, D.R., Raczka, B., Frankenberg, C., Köhler, P., Cheng, R., Smith, K.R., Goulden, M.,

Jung, M. and Lin, J.C., 2023. Satellite-based solar-induced fluorescence tracks seasonal and elevational patterns of photosynthesis in California's Sierra Nevada mountains. *Environ. Res. Lett.*, 19(1), 014008. DOI: 10.1088/1748-9326/ad07b4

Mallia, D.V., Mitchell, L.E., Gonzalez Vidal, A.E., Wu, D., **Kunik, L.**, and Lin, J.C., 2023. Can we detect urban-scale CO₂ emission changes within medium-sized cities? *J. Geophys. Res. Atmos.*, 128, e2023JD038686. DOI: 10.1029/2023JD038686

Mallia, D.V., Mitchell, L.E., **Kunik, L.**, Fasoli, B., Bares, R., Gurney, K., Mendoza, D., and Lin, J.C., 2020. Constraining urban CO₂ emissions using mobile observations from a light-rail public transit platform. *Environ. Sci. Technol.*, 54(24), 15613–15621. DOI: 10.1021/acs.est.0c04388

Kunik, L., Mallia, D.V., Gurney, K.R., Mendoza, D.L., Oda, T. and Lin, J.C., 2019. Bayesian inverse estimation of urban CO₂ emissions: results from a synthetic data simulation over Salt Lake City, UT. *Elem. Sci. Anth.*, 7(1), 36. DOI: 10.1525/elementa.375

In Review:

Huo, X., Raczka, B., **Kunik, L.**, Hicke, J.A., Frankenberg, C., Cheng, R., Slaton, M.R., and Lin, J.C. What is the value of assimilating multiple data streams for improving carbon uptake in the western United States? In revision at *J. Geophys. Res. Biogeosci.*

Grants

2025 NASA Future Investigators in Earth and Space Science and Technology — \$50k
Leveraging remote sensing of photoprotective pigments to understand drought, mortality, and carbon dynamics in Western US forests

2025 Wilkes Center for Climate Science & Policy Dissertation Writing Fellowship — \$30k

2023 FLUXNET Secondment Early-Career Program — \$5k
Exploring photosynthetic responses to warming and drying climate shifts in evergreen forest biomes using the semi-arid Yatir forest in Israel as a case study

2022 National Science Foundation Graduate Research Fellowship — \$144k
Diagnosing drought-induced conifer stress in mountains using solar-induced chlorophyll fluorescence from satellites

Workshops and First-Author Presentations

2026

A carotenoid-based vegetation index for tracking photosynthetic phenology and drought response over North America [Poster] Wilkes Climate Summit, Salt Lake City, UT.

2025

Monitoring photosynthetic responses of wildfire and bark beetle-induced tree mortality using satellite observations of solar-induced chlorophyll fluorescence (SIF) [Oral] American Geophysical Union Fall Meeting, New Orleans, LA.

Mapping conifer forest drought response with photoprotective pigment dynamics across North America using the MODIS Chlorophyll-Carotenoid Index (CCI) [Oral] American Geophysical Union Fall Meeting, New Orleans, LA.

2024

What is the utility of satellite-based solar-induced fluorescence (SIF) for predicting drought-induced forest mortality in the Western US? [Poster] American Geophysical Union Fall Meeting Innovation Session, Washington, DC.

2023

Satellite-based solar-induced fluorescence tracks seasonal and elevational patterns of photosynthesis in California's Sierra Nevada mountains. [Oral] AmeriFlux Annual Meeting, Garner, MA.

Early-career remote sensing workshop instructor. AmeriFlux Annual Meeting, Garner, MA.

Satellite-based solar-induced fluorescence tracks seasonal and elevational patterns of photosynthesis in California's Sierra Nevada mountains. [Poster] American Geophysical Union Fall Meeting Innovation Session, San Francisco, CA.

2022

Multi-scale constraints towards understanding conifer forest carbon uptake in complex terrain within the Carbon Monitoring System-Mountains Project. [Poster] American Geophysical Union Fall Meeting, Chicago, IL.

Participant in Fluxcourse: two-week early-career workshop focusing on foundations of land-atmosphere flux measurement, modeling and synthesis. University of Colorado Mountain Research Station, Niwot Ridge, CO.

2016

Multiscale observational constraints on CO₂ flux estimation using an inverse modeling approach. [Poster] University of Wisconsin–Madison Atmospheric, Oceanic and Space Sciences Poster Reception, Madison, WI.

2015

Multiscale observational constraints on CO₂ fluxes. [Oral] NOAA Office of Education Science and Education Symposium, Silver Spring, MD.

Awards and Honors

- Future Investigators in NASA Earth and Space Science and Technology (FINESST) Fellow — 2025
- Graduate Fellow, Wilkes Center for Climate Science & Policy, University of Utah — 2025
- FLUXNET Secondment Award Recipient — 2023
- National Science Foundation Graduate Research Fellow — 2022
- Distinction in Computer Sciences & Computer Engineering Majors, UW–Madison — 2016

- U.S. DOE Science Undergraduate Laboratory Internship (SULI) Scholar — 2016
- Best Oral Presentation Award, NOAA Student Science and Education Symposium — 2015
- Meyerhoff Undergraduate Excellence Award for Leadership, Service, and Scholarship — 2015
- NOAA Ernest F. Hollings Undergraduate Fellow — 2014

Technical Skills

Statistical and Geospatial Data Analysis:

- **Python:** Jupyter, scipy, scikit-learn, xarray, xESMF, geopandas, cartopy, rasterio, rioxarray, dask
- **R:** tidyverse (dplyr, ggplot2, tidyr), sf, ncdf4, raster, sp, rgdal
- **ArcGIS, QGIS:** Raster and vector data visualization

High Performance Computing & Data Management:

- Advanced Linux/Unix environment expertise including shell scripting
- SLURM job scheduling and monitoring; parallel batch processing
- Automated backup systems (cron jobs, rclone)
- Large-scale dataset management across distributed storage systems (Google Drive, Box)
- Software development with AI tools: Copilot, OpenAI, Claude
- Git version control and repository management

Earth Science Data Products, Models, and Archives:

- **Satellite remote sensing:** MODIS, VIIRS, SRTM, OCO-2/3 (NASA); Landsat (USGS); S5P/PAL, ESA-CCI (ESA); GCOM-SGLI (JAXA)
- **Climate reanalysis:** SNODAS, GRIDMET, TerraClimate, ERA, nClimGrid, SPEIbase
- **Land surface characteristics:** NLCD, ESA CCI Land Cover, SRTM and NOAA ETOPO DEMs
- **Biogeochemical model-data fusion:** FLUXCOM, MODIS GPP, CarbonTracker, solar-induced fluorescence products (GOSIF, CSIF, SIF-ESDR, RTSIF)
- Proficient in data acquisition through USGS LP DAAC, NASA AppEEARS, and Google Earth Engine
- Experience producing datasets using NetCDF Climate and Forecasting (CF) metadata standards
- Community Land Model-5.0 and Community Earth System Model-2: proficiency running and analyzing model output
- **Atmospheric dispersion modeling:** WRF-STILT, HYSPLIT, AERMOD

Additional Methods and Tools:

- Eddy covariance flux data processing: partitioning of ecosystem fluxes through REdDyProc (R) and ONEFLUX processing routines
- **Advanced statistical methods:** Bayesian inverse modeling (prior construction, footprint-based Jacobian/sensitivity matrix assembly, spatial/temporal error-covariance specification, model-data mismatch matrices, Observing System Simulation Experiment validation), geospatial bootstrapping, multivariate kernel density estimation, generalized linear/additive models (GLM/GAM), spatiotemporal analysis and resampling techniques
- **Machine learning / deep learning:** TensorFlow/Keras (custom RNN, stacked LSTM, 1D CNN with attention layers), PyTorch, hyperparameter tuning (keras_tuner), model explainability (LIME), random forests
- **Air quality regulatory modeling & compliance:** AERMOD atmospheric dispersion modeling, EPA HEM-3 human exposure/risk assessment, CEQA and AB 2588 Air Toxics Hot Spots compliance analysis, CalEEMod/EMFAC/OFFROAD emissions quantification, AP-42 emission factor methods
- Additional programming/analysis experience: Excel, JavaScript, MATLAB, Android Studio, HTML, Java, C

Professional Experience

- Postdoctoral Researcher**, University of Utah, Salt Lake City, UT 2026–present
- Developed and validated a gap-filled, gridded vegetation index dataset for North America from NASA MODIS surface reflectance data, including QC and uncertainty characterization.
- Graduate Research Assistant**, University of Utah, Salt Lake City, UT 2021–2026
- Conducted multi-scale analyses to quantify forest carbon cycle responses to drought and disturbance using satellite remote sensing, climate reanalysis, and surface meteorological observations.
 - Communicated findings via first-author peer-reviewed publications and talks at national conferences.
 - Organized and led a cross-departmental working group on vegetation-climate interactions.
 - Secured competitive federal and institutional research funding (NASA, NSF, University of Utah).
- Air Quality Consultant**, Ramboll Group, Salt Lake City, UT 2019–2021
- Developed air quality and greenhouse gas emissions inventories and operational recommendations for clients in chemical manufacturing, data center, airport, and housing industries.
 - Conducted atmospheric dispersion modeling to estimate emissions impacts for regulatory compliance on commercial and industrial development projects.
 - Performed on-site monitoring of air contaminants to verify emissions compliance.
- Research Associate**, Land-Atmosphere Interactions Research Group, University of Utah, Salt Lake City, UT 2017–2019
- Assessed greenhouse gas inventories using surface CO₂ observations, atmospheric models, and Bayesian inverse techniques to constrain CO₂ emissions and uncertainty estimates in the Salt Lake Valley.
 - Designed and taught workshops on Bayesian inverse modeling techniques for researchers at academic and federal institutions.
- Secondary Education Math Teacher**, U.S. Peace Corps, Lesotho 2018
- Taught high school math in a rural district of Lesotho (sub-Saharan Africa) using English and Sesotho.
- Contract Researcher**, NOAA Earth System Research Laboratory, Boulder, CO 2017
- Assessed network densities and meteorological drivers for data assimilation studies using aircraft and in-situ CO₂ data.
- Electrical Engineering Intern**, National Renewable Energy Laboratory, Golden, CO 2016
- Developed a custom control board and solid-state relay configuration for residential “smart” appliances.
- Research Assistant**, Desai Ecometeorology Lab, Dept. of Atmospheric Sciences, University of Wisconsin, Madison, WI 2015–2016
- Integrated ground-based spectrometer data into regional Bayesian inversion techniques to constrain upper tropospheric carbon fluxes.
- Hollings Fellow**, NOAA Earth System Research Laboratory, Boulder, CO 2015
- Performed synthetic data analyses of Bayesian statistical modeling techniques to aid in regional CO₂ flux model development.
- Chainsaw Crew Member**, Rocky Mountain Conservancy, Grand Lake, CO 2014
- Felled hazardous beetle-kill trees near high-use recreation areas in response to Mountain Pine Beetle outbreak.
- Lab Member**, Holloway Air Quality Research Group, University of Wisconsin, Madison, WI 2013
- Examined trends in air pollution and temperature extremes over cities in the US Midwest.

Teaching and Mentoring Experience

- Adult English Language Tutor, Guadalupe School, Salt Lake City, UT — 2024–present

- Software programming mentor, REALM – Research Experience for Undergraduates (REU) program, University of Utah, Salt Lake City, UT — 2022
- Instructor, U.S. Peace Corps Math Education Sector, Lesotho — 2018
- Instructor, Bayesian Inverse Modeling Workshop, University of Utah, Salt Lake City, UT — 2017
- Instructor, Bayesian Inverse Modeling Workshop, NOAA Earth System Research Lab, Boulder, CO — 2017

Professional Service and Volunteerism

- Manuscript referee: Global Change Biology, Science of the Total Environment, Geophysical Research Letters, Environmental Research Letters, Forest Ecology and Management
- IOP Trusted Reviewer status
- Software programming and English language mentor volunteer, Smithsonian Tropical Research Institute, Gamboa, Panama — 2023
- Mentor, City of South Salt Lake Promise Program, South Salt Lake, UT — 2020
- Mentor, Madison Middle School Science Symposium, Wright Middle School, Madison, WI — 2016
- Vice President and Project Manager, Engineers without Borders, University of Wisconsin, Madison, WI — 2015–2016
- Education Outreach Coordinator, Engineers without Borders, University of Wisconsin, Madison, WI — 2014
- STEM Education Volunteer, Madison Children’s Museum, Madison, WI — 2014
- Mentor, Science Olympiad, Edgewood High School, Madison, WI — 2013–2014

Professional Organizations and Certifications

- Member, American Geophysical Union — 2022–present
- Member, Air & Waste Management Association, Great Basin Chapter — 2019–2020
- EPA Method 9 Visible Emissions Certification (through AeroMet Engineering) — 2020

Languages Spoken

English (native), Spanish (conversational)

Personal Interests and Accomplishments

- Long distance hiking: completion of Superior Hiking Trail (2015), Pacific Crest Trail (2017), Arizona Trail (2018), Hayduke Trail (2021), Wind River High Route (2022), and Great Divide Trail (2026)
- Cross country skiing: 12-time finisher of American Birkebeiner (Cable, WI)
- Curling Instructor at the Utah Olympic Oval, Kearns, UT (2018–2022)
- Hobbies: trail running, mountain biking, canoeing, backpacking, cross-country skiing, backcountry skiing, climbing, bicycle touring, canyoneering, classical piano, gardening, woodworking